CS 5045: Computation for the Data Sciences

Practice Midterm Exam

Modules 1–7 • Duration: 60 minutes • Total Points: 100

Part 1: Code Trace (24 Points)

Write exactly what each short program prints. If a program throws an error, write what printed before the error, then state that it is an error and explain why.

Q1.

```
scores = [65, 80, 45, 90, 72]
total = 0
for s in scores:
    if s >= 70:
        total += 10
print(total)
```

Q2.

```
def check_speed(speed):
    if speed > 70:
        return "Ticket"
    elif speed > 55:
        return "Warning"
    else:
        return "Safe"

print(check_speed(60))
print(check_speed(50))
```

Q3.

```
item = {"name": "Laptop", "price": 999.99, "in_stock": True}
print(item["price"] < 500)
print(type(item["in_stock"]))
```

Q4.

```
user = {"username": "data_sci", "level": 2}
user["level"] = 3
user["is_admin"] = False
print(user["level"])
print(len(user))
```

Q5.

```
def calc_bmi(weight_kg, height_m=1.75):
    return round(weight_kg / (height_m ** 2), 1)

print(calc_bmi(70.0))
print(calc_bmi(70.0, height_m=2.0))
```

Q6.

```
def compute_square(n):
    return n * n

def display_square(n):
    print(n * n)

x = compute_square(4)
y = display_square(4)
print(x)
print(y)
```

Q7.

```
class Device:
    def __init__(self, brand, battery_pct):
        self.brand = brand
        self.battery_pct = battery_pct

    def needs_charge(self):
        return self.battery_pct < 20

d1 = Device("Apple", 15.5)
d2 = Device("Samsung", 85.0)
print(d1.needs_charge())
print(d2.needs_charge())
print(d1.brand)
```

Q8.

```
class Plan:
    def __init__(self, client, base_rate):
        self.client = client
        self.base_rate = base_rate

    def final_cost(self):
        return self.base_rate

class PremiumPlan(Plan):
    def __init__(self, client, base_rate, concierge_fee):
        super().__init__(client, base_rate)
        self.concierge_fee = concierge_fee

    def final_cost(self):
        return self.base_rate + self.concierge_fee

p1 = Plan("John", 100)
p2 = PremiumPlan("Sarah", 100, 50)
print(p1.final_cost())
print(p2.final_cost())
print(p2.client, p2.concierge_fee)
```

Part 2: Types and Representation (24 Points)

Q9. Write the exact Python data type (int, float, str, bool) for each of the following expressions:

- False
- "True"
- 42
- 42.0
- "-1"

Q10. Recall the baseline habit of the **four-line audit** run on every new DataFrame (df). Write out the four specific commands and a brief phrase explaining what data dimension or problem each line inspects.

Q11. A column named transaction_amount should contain purely decimal values (e.g., 12.50, 99.00), but one human entry was recorded as $15.00. After loading the data with pd.read_csv(), what type will Python assign to this column, and name one quantitative analysis step that will fail because of it.

Q12. A file loaded via csv.DictReader parses a row as {"item": "Desk", "price": "150", "tax_exempt": "False"}.
- Write one line of Python to safely extract and convert price into an integer.
- Write one line of Python to safely convert tax_exempt into a proper boolean representation.

Q13. If a series contains a boolean column where True marks a customer who downgraded their subscription, and False marks someone who kept it, what distinct insights do .sum() and .mean() tell you about your subscription base?

Part 3: The Five Operations (8 Points)

Circle or write down the primary computational framework operation (Represent, Filter, Transform, Aggregate, Communicate) that matches each scenario.

Q14. df = pd.read_csv("hospital_records.csv")

Q15. Classifying credit scores into categorical tiers: "Poor", "Fair", "Good", "Excellent" without dropping rows.

Q16. df.groupby("department")["salary"].mean()

Q17. Isolating all customer profiles where age >= 65.

Q18. Drafting an executive summary paragraph containing key insights to hand off to the regional manager.

Part 4: Write Code by Hand (25 Points)

Write clean, logical Python snippets. Minor syntax blunders (like a missing colon) are fine if structural intent is clear.

Q19. Given a list salaries = [95000, 120000, 85000, 105000, 110000], write a for loop that prints only the salaries that are strictly greater than 100000.

Q20. You have a list of employee dictionaries called staff:

```
staff = [
    {"name": "Alice", "sales": 15000, "years": 3},
    {"name": "Bob", "sales": 8000, "years": 1}
]
```

Write a loop that updates each dictionary in place by adding a new key "bonus". If sales >= 10000 and years > 2, set "bonus" to 1000. If sales < 10000, set "bonus" to 0. Otherwise, set "bonus" to 500.

Q21. Given a list of transaction row dictionaries:

```
transactions = [
    {"category": "Food", "amount": 12.50},
    {"category": "Tech", "amount": 150.00},
    {"category": "Food", "amount": 8.25}
]
```

Write a script by hand to **group the amounts by category** into a dictionary of lists, and then calculate and print out each category alongside its mathematical average. *Do not use pandas.*

Q22. Look at the following parent class:

```
class Employee:
    def __init__(self, name, base_salary):
        self.name = name
        self.base_salary = base_salary

    def calculate_pay(self):
        return self.base_salary
```

Write a subclass called Manager that adds an attribute called bonus. Use super().__init__ to handle the initialization of name and base_salary, and override the calculate_pay method to return the sum of the base salary and the bonus.

Part 5: Short Answer (19 Points)

Provide concise explanations (2–4 sentences long).

Q23. In object-oriented programming, how does a single class method handle calls from two distinct objects with different data and yield unique outcomes despite having only one block of code written? Reference the underlying mechanics.

Q24. Define what a **proxy variable** means within data science ethics. Explain why removing a protected characteristic (like race or gender) from a data representation layer frequently fails to stop a model from making biased predictions.

Q25. A team needs to answer a prompt from a manager: *"Do remote workers take more sick days than in-office workers?"* Describe the technical execution steps you would follow in chronological order, explicitly labeling the computational operation matching each step.